Performance Testing & Acceptance Report

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	5 Marks

Performance Benchmarks & Load Verification:

Evaluates system behavior, API response times, and stability under peak traffic conditions.

Test Metric	Target SLA	Observed Benchmark	Evaluation Result
Groq AI Generation Latency	< 3,000 ms	980 ms - 1,450 ms	**Exceeds Target (Passed)**
Frontend CDN TTFB (Vercel)	< 500 ms	120 ms - 240 ms	**Passed**
Database Query (Asyncpg Pool)	< 100 ms	45 ms average	**Passed**
Backend API /health Endpoint	< 200 ms	65 ms	**Passed**
Concurrent Users Capacity	200+ users	500 concurrent connections	**Passed**
JSON Parsing Reliability	99.9% valid	100% valid (0 truncations)	**Passed**